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Introduction

Understanding Object Representations in the Brain

Hebart and colleagues¹ used a data-driven approach to identify core dimensions of object representation, then linked them to brain activity². These findings show how the visual system supports object recognition by mapping perceptual features to semantic knowledge.

However, visual and semantic dimensions are often heavily intertwined.

Research Question

1. How do word- and image-based behavioral representations differ?
2. How do these representations relate to fMRI responses to images?

Behavioral experiment

- Stimuli: 1,388 diverse, unambiguous object words, sampled from the THINGS database (figure 2).
- Methods:
 - Triplet task using large-scale online crowdsourcing (1.3 million similarity judgments).
 - Embedding obtained from SPoSE model¹, trained to capture human similarity choices.

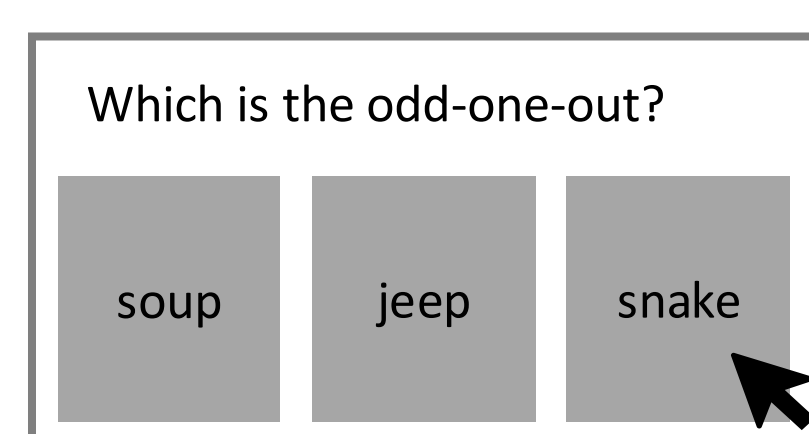


Fig 1. Triplets task

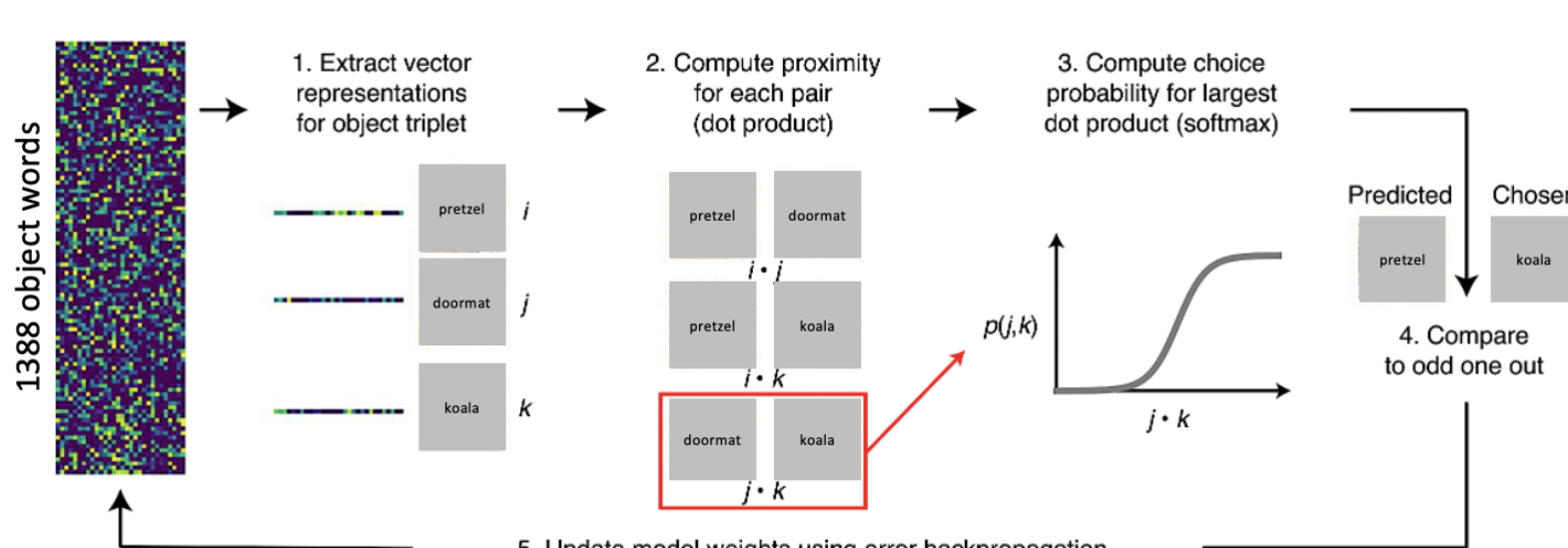


Fig 2. SPoSE model workflow

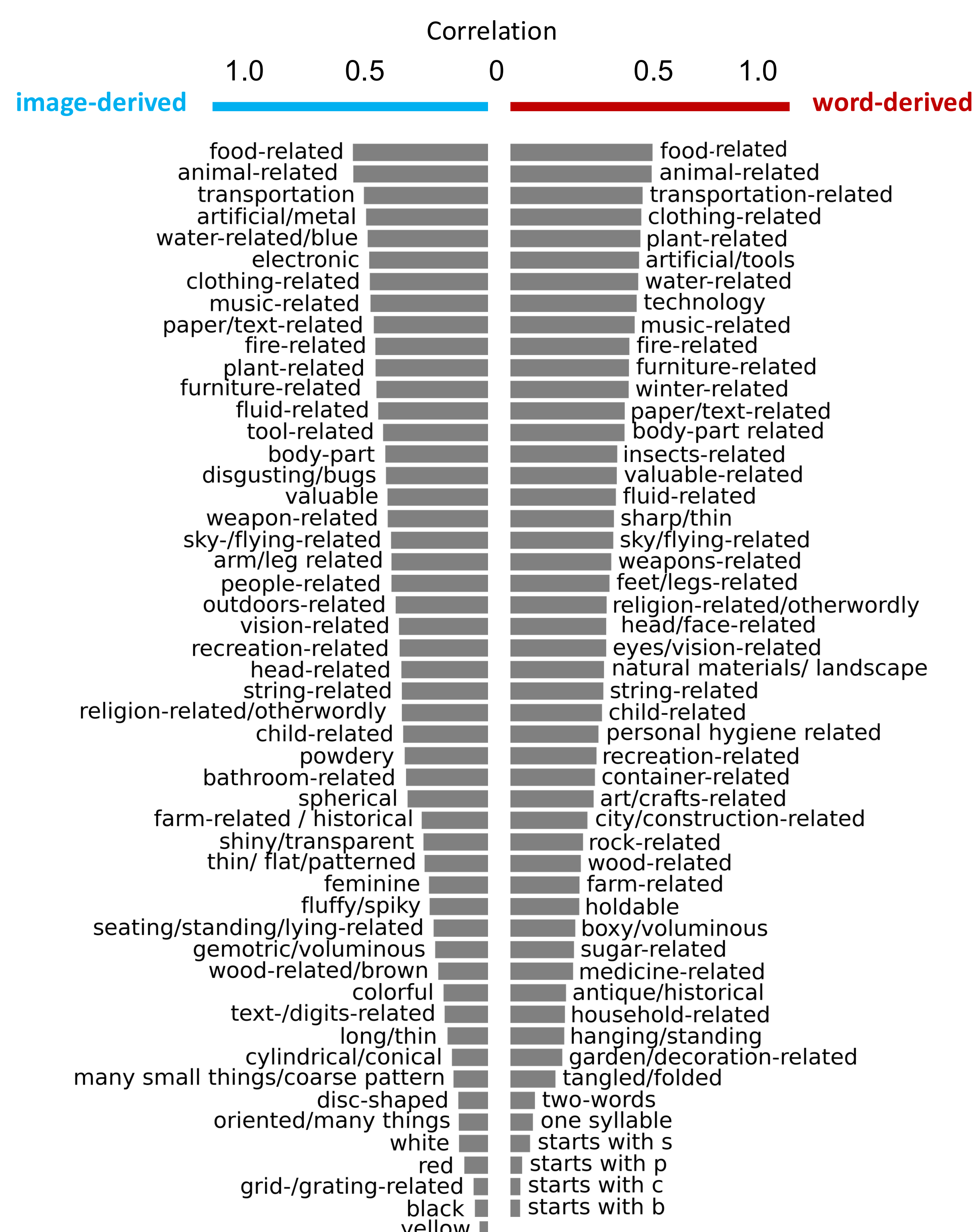


Fig 3. Correlations of word-derived dimensions with any image-derived dimensions and vice versa → color and texture dimensions are unique to image representations



Laptop

Fig 4. Rose plots illustrating the representation of 'laptop' in both images and words.

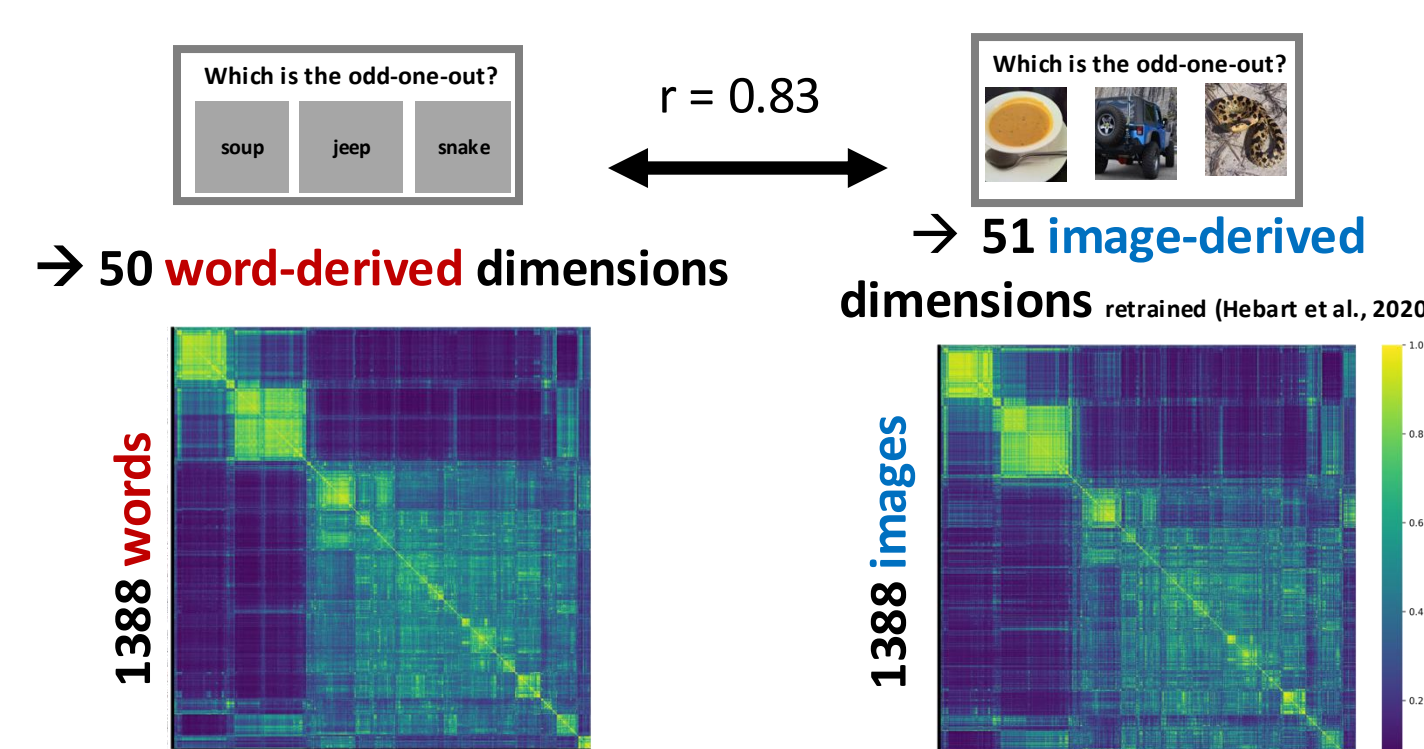


Fig 5. The high correlation between word-derived and image-derived dimensions.

fMRI Study Design

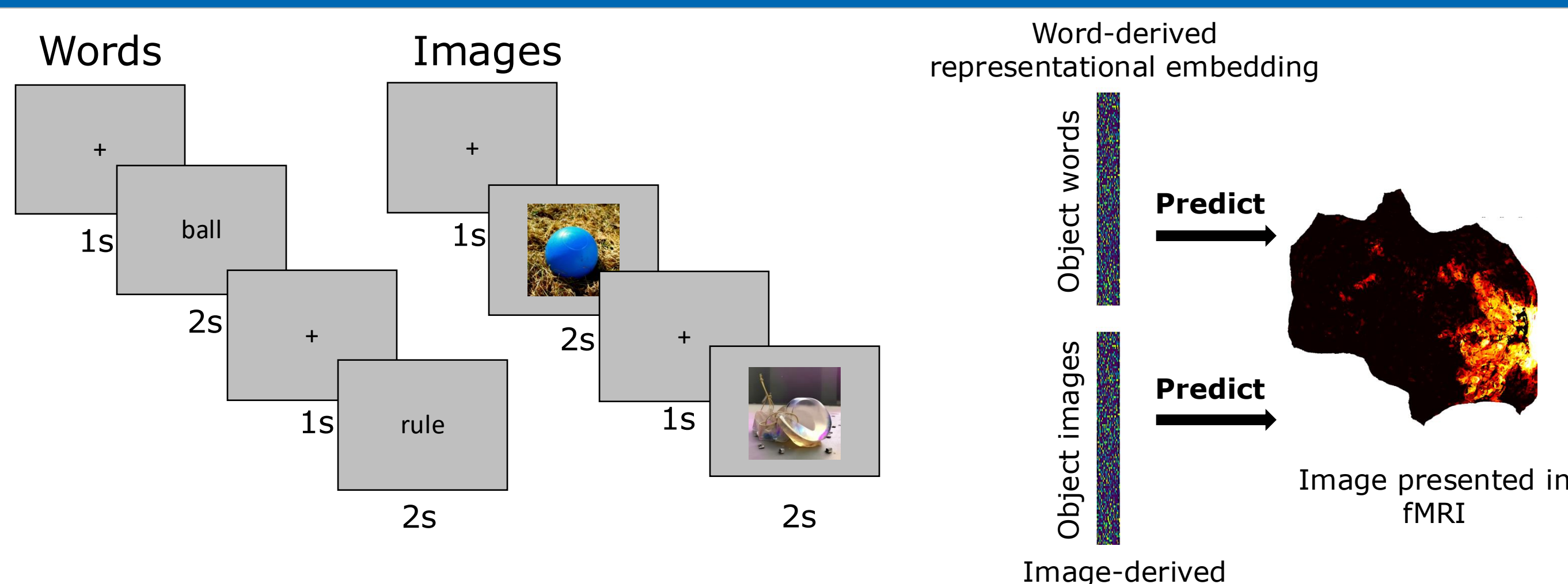


Fig 6. Design. fMRI dataset (n = 5; 15 sessions). Words: 480, repeated 8x. Images: 960 pairs, repeated 4x. Each word paired with one THINGS+ and one dissimilar THINGS image¹.

Fig 7. Analysis workflow

fMRI Results

Model performance for image-based fMRI stimuli

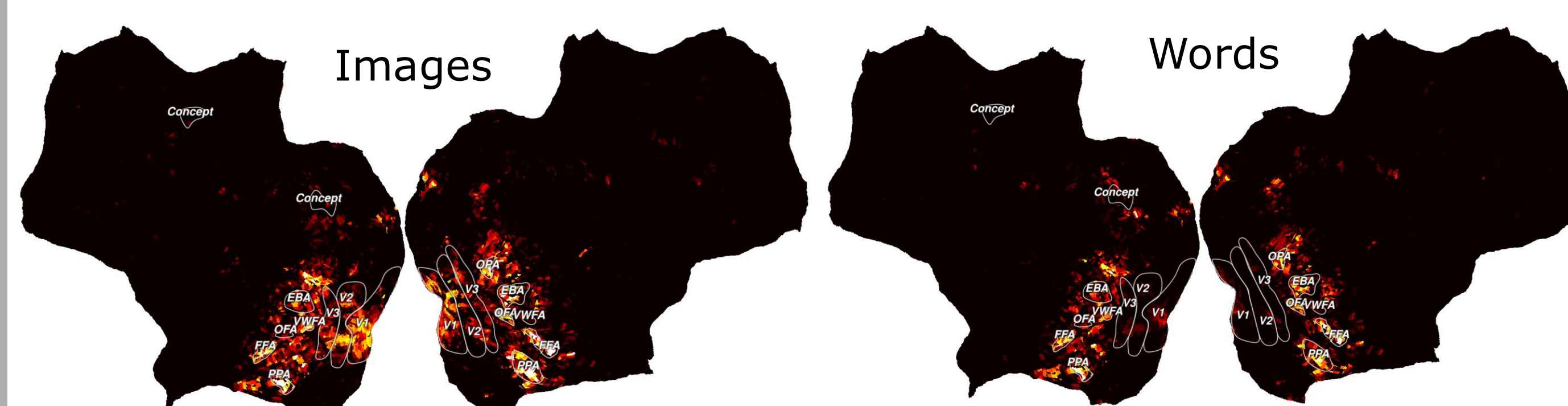


Fig 8. Model performance shows that both image-derived embeddings and word-based models accurately predict neural responses during image presentation.

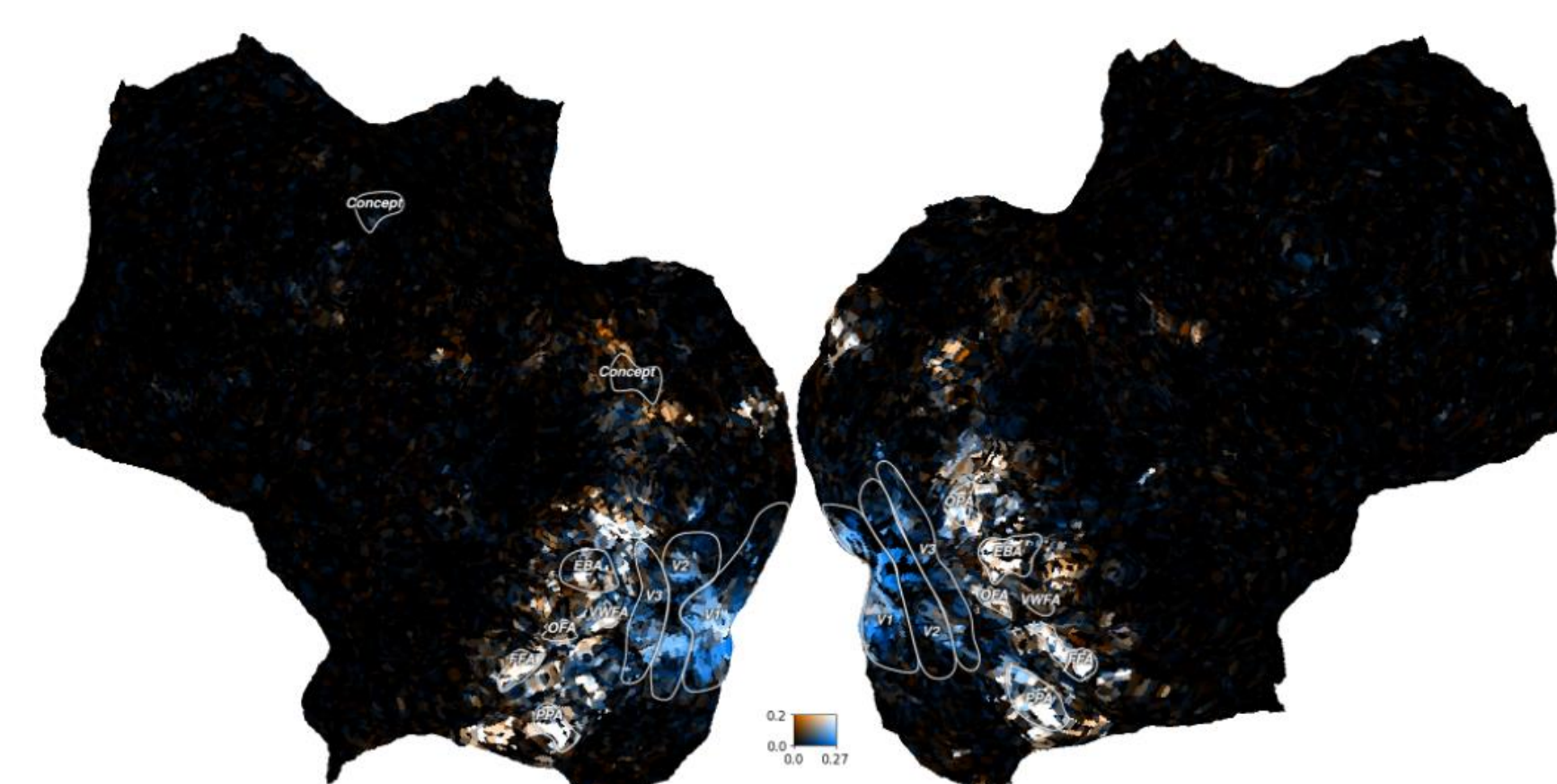


Fig 9. Shared representations (white) are concentrated in the ventral temporal cortex, highlighting its role in multimodal representation. Word-only (orange) extends into right frontal regions, whereas image-only responses (blue) are confined to visual areas.

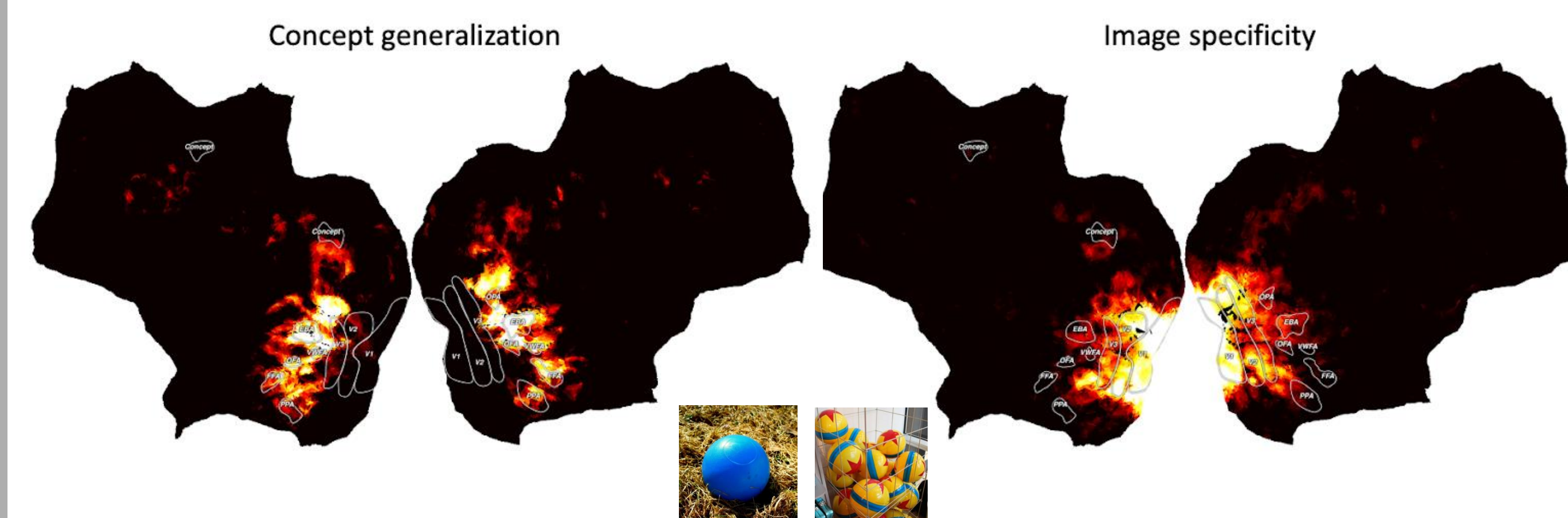


Fig 10. Concept generalization and image-specific representations in the human brain. Image exemplars for the concept of 'ball'.

Conclusion

Behavioral overlap: Word embeddings align with object image dimensions, but capture mainly semantic information (minimal shape, color, texture).

Neural overlap & distinction: Word-derived and image-derived embeddings show partially overlapping representations in ventral temporal cortex.

Functional gradient: Along the visual hierarchy, early visual cortex preserves visual/exemplar-specific features, while higher-level regions encode semantic/conceptual information.

Conclusion: Evidence for a shared high-level semantic space across modalities.

References

1. Hebart, M. N., Zheng, C. Y., Pereira, F., & Baker, C. I. (2020). Revealing the multidimensional mental representations of natural objects underlying human similarity judgements. *Nature Human Behaviour*, 4(11), 1173-1185.
2. Contier, O., Baker, C.I. & Hebart, M.N. Distributed representations of behaviour-derived object dimensions in the human visual system. *Nat Hum Behav* 8, 2179–2193 (2024).